

HRISHIKESH BELAGALI

Undergraduate Student

📍 East Lansing

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SUMMARY

Undergraduate computer science major with research interests in reservoir computing, reinforcement learning, molecular dynamics, high performance computing, and quantum algorithms.

SKILLS

Machine learning, deep learning, Monte Carlo optimization, quantum algorithms, numerical methods.

Languages: Python, C, C++, $\text{\LaTeX}$

Packages: NumPy, PyTorch, scipy, Qiskit, PyEMMA, hyperopt, VMD-Python

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S)

Expected Graduation: 2027

Current GPA: 4.0

Michigan State University Honors College

EXPERIENCE

2024–

Undergraduate Research Assistant

Vermaas Lab, Plant Research Laboratory

- Investigated the interaction of the chloroplast targeting protein (At4g16060) with thylakoid membranes using molecular dynamics.
- Analyzed the effect of PDLIP-5 (Plasmodesmata Located Protein) on water flow rates through PIP1:4 aquaporin tetramer using grid-steered molecular dynamics.
- Calculated protein transition kinetics using Markov State Models (MSMs) and transition path theory.

PROJECTS

Python
NumPy

aqua-blue

github.com/pypi.org

Secondary maintainer of aqua-blue, a package building echo state networks.

- Implemented reservoir state time-lag, sparsity and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated prediction capability by predicting quantum time evolution of Gaussian wave packets.

Python
NumPy
hyperopt

aqua-blue-hyperopt

github.com/pypi.org

Primary maintainer of aqua-blue-hyperopt, a package implementing hyperparameter optimization functionality for aqua-blue models.

- Created modular hyperparameter optimization functionality for aqua-blue reservoir computers.
- Implemented custom loss function capabilities for flexible model evaluation.

Python
PyTorch
pytest

neurop

github.com/pypi.org

Primary maintainer of neurop, a package implementing neural operators.

- Developed Fourier Neural Operator (FNO) and DeepONet architectures for solving PDEs.
- Implemented the Fractional Fourier Transform for use in Complex Neural Operators (CoNO).
- Built CI/CD pipeline with GitHub Actions for automated testing and deployment.

Python
Qiskit

Discretized Quantum Adiabatic Evolution

github.com

Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques

Python
Qiskit

Quantum Arithmetic Circuits

github.com

Built quantum circuits to perform addition, subtraction and multiplication using the Quantum Fourier Transform.

Python
OVITO
pyMOL

Genetic Annealing to Determine Protein Structures

github.com

Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irback's off-lattice model energy equation ($\text{RMSD} < 3.0$).

Python

Quadratic Unconstrained Binary Optimization

github.com

Stochastic tunneling-enhanced simulated annealing for solving QUBO-type problems.

Python
NumPy

Monte Carlo Integration

github.com

Monte Carlo integration with importance sampling to numerically evaluate complex integrals.

Python
NumPy

Langevin Monte Carlo

github.com

Metropolis-adjusted Langevin algorithm for sampling from intractable probability distributions.

CONFERENCES

2025
PRL, MSU

Aquaporin Water Transit Modulation via Protein Capping
Submitted abstract to ACS Fall 2025 Conference

acs.digitellinc.com

2025
PRL, MSU

Evaluating the Interaction of Chloroplast-targeting Protein (At4g16060) with Thylakoid Membrane
Submitted abstract to First Great Lakes Plant Science Conference

HONORS AND AWARDS

- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)
- MSU President's Scholarship (2024)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)